Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/30/2007
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Crop and Bio Mass Production		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Beyl, Caula (Alabama A & M) Mitchell, Cary (Purdue University) Brown, Paul (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Three NSCORT (NASA Specialized Center of Research and Training) projects address the feasibility of producing edible biomass in a space regenerative life-support system. One project addresses a novel approach to significantly reduce the ESM (Equivalent System Mass) of growing food crops in controlled environments; the second project tests the potential to use fungi to degrade non-edible crop residues while generating edible mushrooms; and a third project addresses the feasibility of feeding crop or bioreactor residue to fish that can be eaten. A novel lighting system for higher plants has demonstrated incremental improvement in electrical energy utilization during the project by correcting software glitches, by creating growth compartments that reflect escaped photons back into the crop stand, by planting dense crop stands that close fast, by positioning plants directly in front of one-sided light strips, and by relying upon the "plasticity" of plant growth to seek out light. Appropriate red/blue ratios to maintain this "light magnet" are being defined. Intracanopy lighting is superior to overhead lighting for crop quantum efficiency and electrical-energy utilization once foliar canopies have closed. It has been confirmed that reproductive development occurs normally with intracanopy LED lighting. Remote position sensing of canopy height and automated switching of light engines is in development. Once needed engineering upgrades to the prototype intracanopy lighting system are implemented, ESM for crop lighting should be reduced at least an order of magnitude relative to that of traditional overhead lighting systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Development of solid-state, intracanopy and close-canopy crop-lighting systems will solve the light-distribution problem that has long limited the photosynthetic productivity of overhead-lighted crop stands, and will permit the use of low-power light sources. These advances will enable year-round, warehouse-based, controlled-environment agriculture (CEA) production of salad greens, herbs, and certain vegetables to be profitable in local markets, especially if coupled with sources of cheap, bio-derived energy, and competitive with produced trucked from distant production sites. 2. Developing methods for maximum crop yield at minimum cost in controlled-environment agriculture (CEA) 3. Developing genetic engineering of CEA crops for superior performance
Task Progress:	BioMass Production: LED (Light Emitting Diode) Lighting Project: Simultaneous intracanopy vs. overhead lighting configurations. Following five trials with the first LED array in either the intracanopy (IC) or overhead (OH) configuration, the original array has been maintained in the OH configuration, and the second prototype is in the IC configuration. In the interest of determining the potential of LEDs for crop production involving reproductive development, a side-by-side IC vs. OH comparison using 'Triton' pepper was conducted. After almost one month of growth, the plants showed severe intumescence symptoms, and at that time the LED settings were adjusted to provide 15% of incident PAR from the blue LEDs. This adjustment did not mitigate the intumescence symptoms with pepper. Harvesting continued and results showed that flowers and fruit form and ripen normally under both lighting conditions, in spite of intumescence. We conducted further experimentation with 'Triton' pepper, to determine if the addition of UV-A radiation in the form of "blacklight" can mitigate the development of these tumor-like structures. Tests were conducted with both intracanopy and overhead LEDs, and each array growth compartment is divided into four chambers, three with differing durations of UV-A light and one without UV. In the overhead LED system, blue levels were set at 50% of 300 µmol/m2/s of total photon flux. In the intracanopy system, 90% blue was attempted, but the blue power with only one light engine energized allowed only about 40% blue to be obtained. UV-A black lights were installed in each growth compartment so that one was energized during the 16-h day, one was energized during the 8-h night, and one was energized continuously. The fourth compartment was not supplemented with UV light. The results of this test were inconclusive. UV light may have reduced intumescence, but with the limited amount of UV present it was difficult to discern. In order to evaluate this more clearly, three plants of a rapidly growing tomato cultivar were grown within each light system. The experimental design called for using two UV black lights on one plant, one on another plant, and none on a third, while maintaining wide spacing between the plants. Unfortunately, this cultivar of tomato showed no indication of oedema under red and blue LEDs, and the experiment was terminated. LED Lighting Project: HELIAC SBIR. A phase-2 SBIR to the Orbital Technologies Corporation (ORBITEC) to develop an automatic canopy-detection / light-switching system (High efficiency lighting with integrated adaptive control or HELIAC) utilized cuttings of tomato and sweetpotato plus cowpea seedlings. Results of initial testing with each species providing different canopy architectures were reported to ORBITEC. After baseline conditions were established a cowpea stand was grown from seed until complete canopy closure and different HELIAC settings were tested throughout the life cycle. Further testing of the HELIAC system occurred with a single cowpea stand divided into separate compartments within which the floor, walls, and ceiling were lined either with a reflective white film or a matte-black, light-absorptive felt. At the same time the HELIAC hardware was separated into two independently operating systems. The HELIAC response to the surfaces and the canopy architectures indicated that the white surfaces provided more light to the plants while the black surfaces allowed better accuracy of position detection. The HELIAC system has been reconfigured into an overhead array of one system and lettuce plants are being grown beneath this LED array in a close-canopy configuration while hardware testing proceeds. All results are reported to ORBITEC on a monthly basis. LED Lighting Project: Canopy Gas exchange. The design of the Minitron 3 gas-exchange cuvette system for intracanopy LED lighting of small crop stands is complete. The final design has been approved and construction has commenced. ALS Candidate Crop Cultivation. Sweetpotato. Plants grown with automatic fertigation formed many more roots than those groqwn with time-release fertilizer. An experiment comparing high-porosity peat-based medium to coarse (Turface) and fine (Profile)-grained calcined clay was showed good root growth under all conditions, but the Profile medium had the most consistent high level of root production while the Turface media had the lowest root production. ALS Candidate Crop Cultivation. Strawberry. New bare-root strawberry plants of cultivar 'Seascape' were planted with ten plants each of this cultivar placed in each of three growth chambers. 'Seascape' was the highest-producing, best-tasting cultivar identified in previous

	experimentation, so the focus of current experimentation is on this cultivar. 'Seascape' was found to be truly day-neutral producing equal fruit yields (number and weight) regardless of photoperiod. The focus of current experimentation is to minimize electrical power for plant-growth lighting by finding the shortest day length possible without impacting plant productivity. In this round of experimentation, the chambers are set for 10-h, 12-h, and 14-h photoperiods with 350 µmol/m2/s fluorescent and incandescent radiation. Plants are pollinated with a vibrating wand and fruit are harvested, weighed, tasted, and rated. There have been several rounds of fruit-quality throughout the summer and autumn. Plants have been pruned back and allowed to regrow to stimulate a new flush of flower and fruit production. ALS Candidate Crop Cultivation. Tomato. Tomatoes were grown for a student studying antioxidant levels in the Mauer Lab in the Food Science department. The first tomato harvest occurred on March 26th, and tomatoes were harvested on a weekly basis thereafter. Tomato cuttings taken from mature vegetative tissue developed into more compact versions of mature plants, and fruit developed on those plants, which were 20% the size of the parent plants. Vegetative tomato waste biomass as well as biomass from sweetpotatoes and strawberries was supplied for use in the PAABLO testing. Fungi used on non-edible crop residues while generating edible mushrooms. Project completed prior to the present reporting period. Annual report submitted in Task Book 2006. Feeding crop or bioreactor residue to fish that can be eaten. Project completed prior to the present reporting period. Annual report submitted in Task Book 2006.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
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